

Gravity: Why It's Not a Force

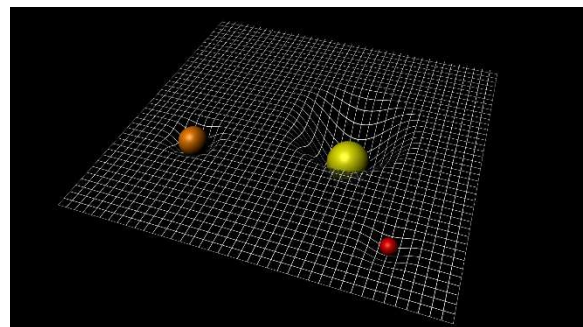
The Apple vs. Elevator

The speed limit of the universe

Newtons law of universal gravitation ($F = G \frac{m_1 m_2}{r^2}$) is the reason why we were able to launch rockets into space and we still use it to this day. However, with this equation Einstein discovered a cosmic glitch; Newtons Equation implies that gravity has infinite speed. If the Sun would suddenly disappear, the gravitational field change would be felt on Earth instantly ($t = 0$). This would create a massive paradox with physics; According to special relativity, the speed of light (c) is the cosmic speed limit for information and causality (You can read more about the speed of light on the document “Speed of Light: Why it’s Not a Target, But a Limit”). In conclusion this means that nothing can travel faster than light; not even gravity. If we would travel from the Earth to the Sun at the speed of light, it would take us around 8 minutes and 20 seconds, meaning that if the sun would disappear Earth would continue its usual orbit for another 8+ minutes around the empty space where the sun used to be located at.

How Einstein solved the Paradox: General Relativity

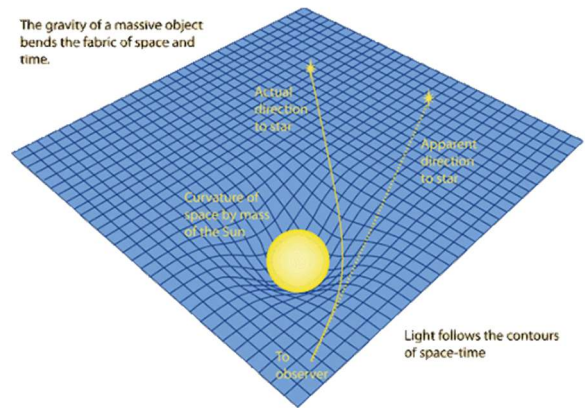
To solve it Einstein started to reimagine what gravity is. He believed that gravity as an instantaneous pull between masses was not the solution but instead proposed that space and time are fused into a four-dimensional fabric which is called “Spacetime”. Spacetime is curved by matter and energy, and the curvature tells matter how to move. The Earth doesn’t get pulled by the sun; it just follows the straightest possible path through a valley of Spacetime made by the suns massive weight. When a massive object moves it creates gravitational waves which is like dropping a stone into water. Einstein’s theory claims that those gravitational waves travel at the speed of light. So only after 8 minutes and 20 second would the Earth fly straight away into the dark cosmos.



Picture 1; How different mass affects Spacetime

Spacetime Geodesic

If we see an apple fall, we see it nearing the ground and falling towards it until it impacts. It looks like a force is pulling it down. However, Einstein turned this upside down; The apple is completely stationary in its own reference (freefall), while the ground itself is accelerating towards it. On a paper (flat surface) the shortest route between 2 points is a straight line. If the surface however is curved the shortest path is a curve (great circle). objects always travel in a straight line in 4D Spacetime according to relativity. Those are called straight paths Geodesic. This means that Mass and energy don't pull objects; they warp the grid of Spacetime around them. When for instance an object orbits Earth, its travelling at a perfectly straight line in 4D Spacetime. However because of the Earths mass this path in Spacetime is bent into a loop resulting in the object falling towards Earth.



Picture 2; How Spacetime affects movement

The Math

$G_{\mu\nu}$ represents the curvature of Spacetime resulting on what energy is present. $\Lambda g_{\mu\nu}$ represents the density of empty space itself, originally added by Einstein to keep the universe static.

$$G_{\mu\nu} + \Lambda g_{\mu\nu}$$

$T_{\mu\nu}$ represents mass, energy, momentum and pressure ($E = mc^2$) in a given region of space. It's the source of gravity. The modifier $\frac{8\pi G}{c^4}$ acts as the conversion factor. The speed of light (c) is raised to the fourth power, which is an astronomically big number and tells us that Spacetime is insanely stiff; a colossal amount of mass is needed to curve it even a bit.

$$\frac{8\pi G}{c^4} T_{\mu\nu}$$

John Wheeler summarized this with a well-known quote: "Spacetime tells matter how to move; matter tells Spacetime how to curve." – John Wheeler. Also note that there's not " F " for force anywhere.

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Engineering: Accelerometers and Inertia

How does an Accelerometer work

A linear accelerometer chip is a mechanical system scaled down to the micro level using MEMS (Micro-Electro-Mechanical Systems). Inside the chip there's tiny proof mass suspended by silicon springs. Placed around it are fixed capacitive plates. When the sensors housing moves the mass wants to stay in place resulting in the electrical capacitance changing between the mass and plates. The chip measures this changes and translates it into acceleration value

The Desk Paradox

If we put a smartphone on a desk, the acceleration should be zero according to Newton because its state is stationary. However, the sensor will read exactly $1g$ (9.81 m/s^2) pointing upward. This happens because according to Einstein the natural state of motion in Spacetime is freefall. The desk is a barrier; its constantly pushing up against the sensor, forcing the proof mass to deflect downward. It reads $1g$ because the desk is violently accelerating the device out of its natural unforced path.

The Freefall proof

If we would now drop the smartphone from high up, the springs inside the MEMS sensor would snap back to the exact centre. It would read exactly $0g$. In Newtonian physics it would be summarized with the object accelerating downwards at 9.81 m/s^2 due to a force. According to Einstein (and the sensor) the sensor is experiencing absolutely zero force and zero acceleration, its resting, coasting along the curvature of Spacetime.

Conclusion

While gravity is not a force, doesn't mean that Newton was wrong; Newtons equations are a mathematically perfect approximation of general relativity when gravity is slow and well below the speed of light (c). You're not sitting on a chair and being pulled down by a force; The chair is stopping your straightest path through Spacetime. Einsteins Theory and equations are an addition to Newtons, they don't make Newton's old equations wrong, they expand them.

Sources

<https://en.wikipedia.org/wiki/Accelerometer>

https://en.wikipedia.org/wiki/Geodesics_in_general_relativity

<https://en.wikipedia.org/wiki/Spacetime>